Image analysis (MicroICS) hands-on session: exercises

Part of: "Deep dive into the statistical approaches used to validate potential standardised laboratory methods"

The golden thread running through MicroICS is a single pattern of analysis, and it applies to your own question just as much as to questions studied on the Training School:
biofilm image → group labels → structural features → model training → what separates the groups → prediction on unknown (treated) images → does a treatment changes the structure

We use eight *Listeria* strains as the worked example, decoding bacterial strain identity from biofilm images. **You can download the sample images here https://portal.ijs.si/nextcloud/s/LC93n9A97Ekme7X.**

But keep asking yourself throughout: how would this map onto my bacteria, my fungus, my favourite microbial pet — my treatment, my question? That translation is the point of the session.

The pattern answers three questions, and each part of the exercise tackles one:

- What distinguishes the groups? (training)
- Does a treatment move the structure — where, and how much? (inference on treated samples)
- Which of those changes is worth chasing in the lab? (the hypothesis)

Objective 1: Turn a biological question into a label

The label is the ground truth that you attach to each image, written into its filename (image) after _st_. Change the label and you change the question; the pipeline never changes.

Do: open the training image folder and look at the filenames. Find the _st_ part. That is the label.

Think and write: How would you name the images to distinguish clinically relevant from non-relevant strains? How would you label images for your question?

Take home: your question becomes a machine learning task the moment you can write its answer into the filename.

Objective 2: Design a dataset that can answer your question - look before you model

Do: open a few 3D images across strains before running anything.

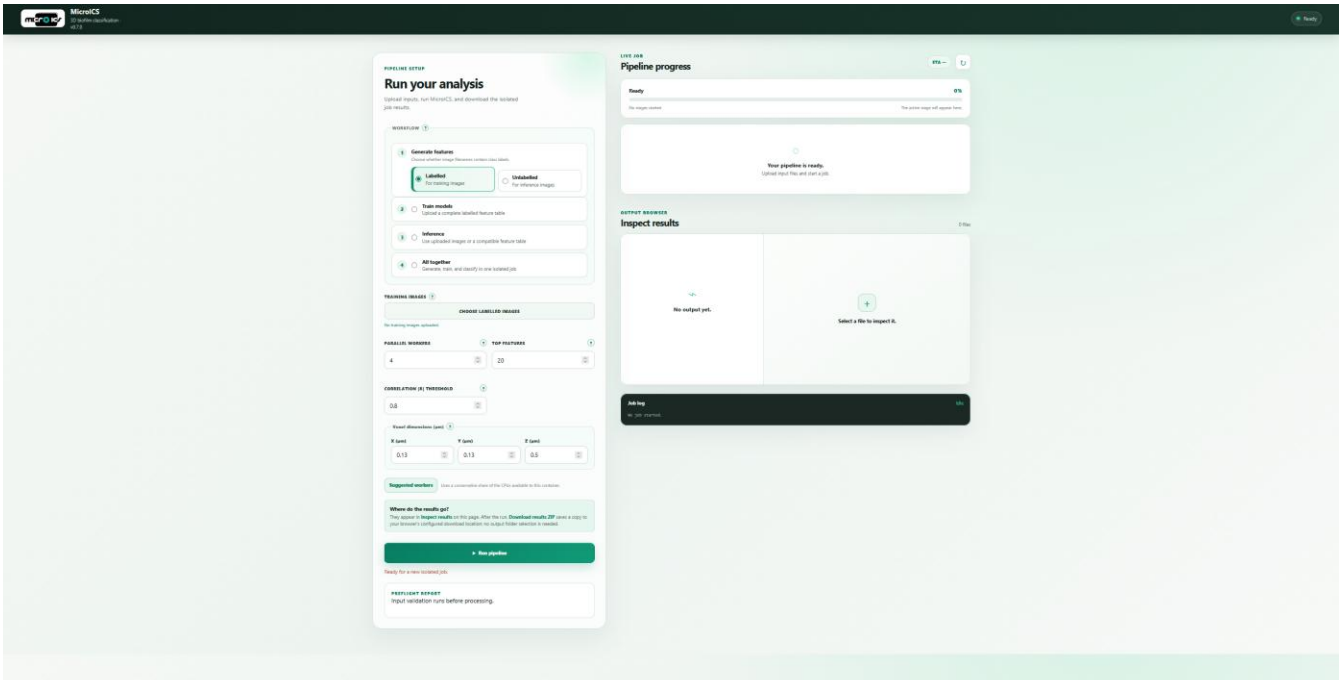
Notice: ATCC 19115 sits dense near the surface; L1764 fills the volume evenly; some sit in between and are genuinely hard to tell apart.

Take home: knowing what is easy and what is hard by eye gives you something to check the model against.

Objectives 3 – 5: Install and run MicroICS independently

Installation

- Open the document "Installation Guide" and start MicroICS (**Restarting the existing container of MicroICS**). The window that opens should look like this:

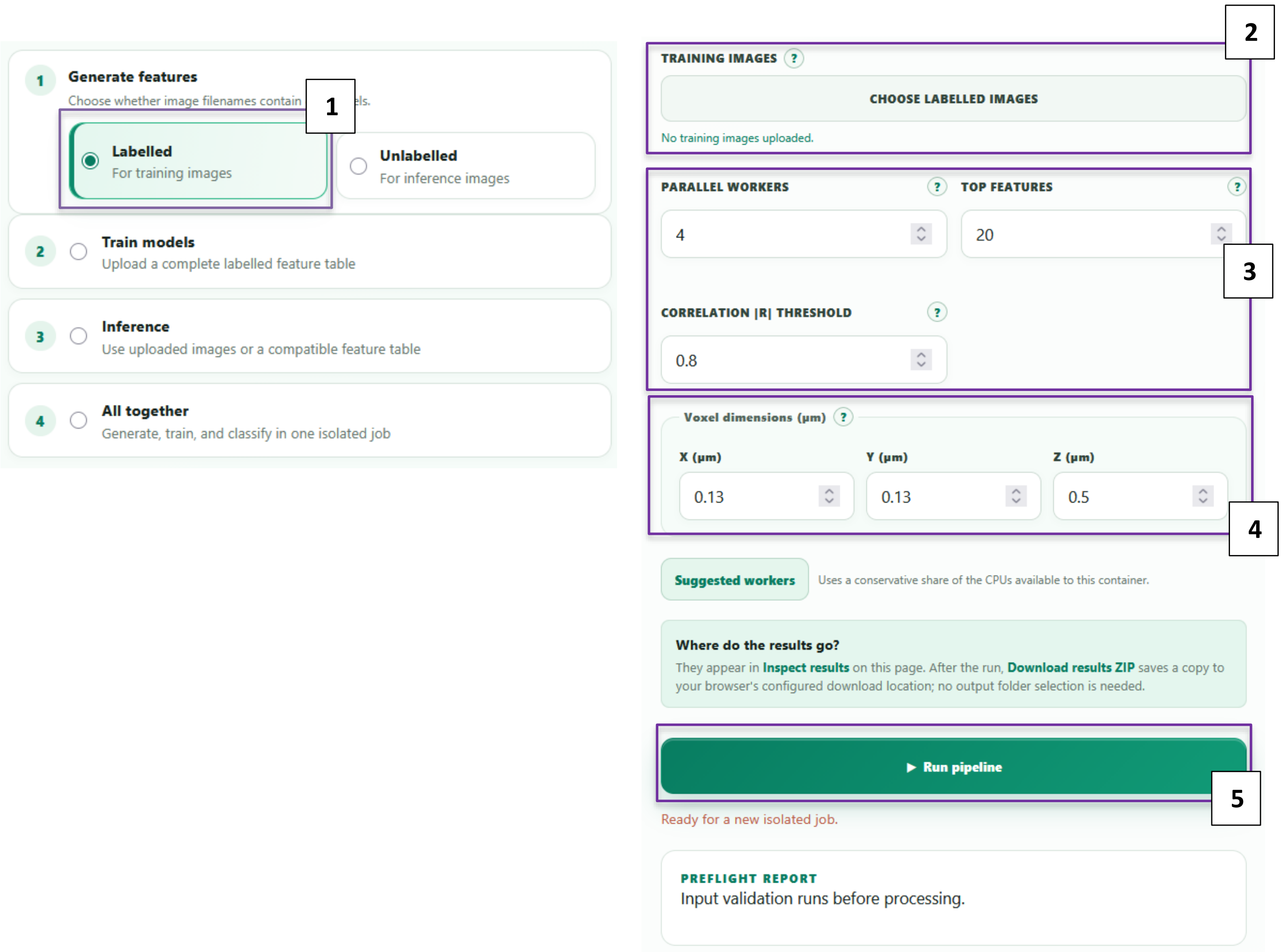


Generate features

Do:

1. Choose “Generate features” in mode “Labelled”. Labelled mode extracts the label from each filename as well, so the features can be used for training.
2. Point the software to the folder containing your images in TIFF format, using “Choose labelled images”.
3. Set the running options. “Parallel workers” – how much of your CPU to dedicate to the analysis: use full capacity when you are not otherwise using the computer, and less if you are working on it at the same time.
   At this point, “Top features” and “Correlation |R| threshold” do not matter.
4. Adjust the voxel dimensions (the pixel size in micrometres) to match your images.
5. Click “Run pipeline”.

When the run starts, the software also checks the image naming and reports the number of classes it found.



See the results: View the results by downloading the ZIP file. Open datafile.tsv – it has one row per image and roughly 2,700 columns, each representing a structural measurement (shape, texture, intensity, or spatial arrangement).

Take home: the model never sees your images. It sees this table. Everything downstream is built on it.
Parameters and every output file are described in the homework document.

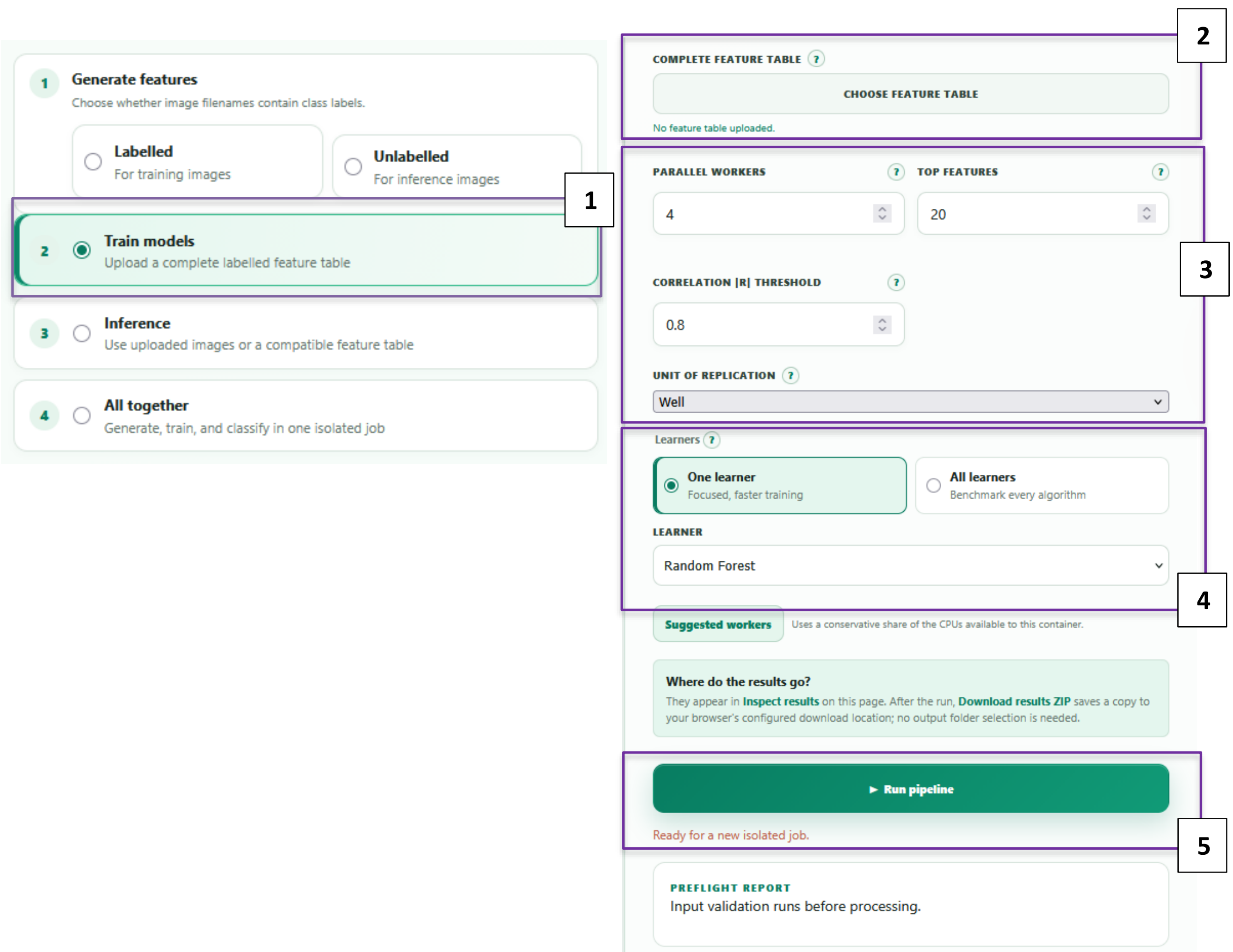
Train a model

We provide a pre-made model for the training school so we do not spend the session waiting. **Skip this part for now, but not for the homework.**

Do:

1. Choose "Train models."
2. Navigate to the folder containing datafile.tsv (referred to here as the feature table) using "Choose feature table".
3. Set the running options:
   "Parallel workers" – how much of your CPU to dedicate to the analysis: full capacity when you are not otherwise using the computer, and less if you are working on it at the same time.
   Correlation analysis – choose how many of the most important features to analyse, and the threshold for the analysis. This helps identify redundant features.
   Unit of replication – specify what counts as an independent experiment (day, plate, well, or position within the well). This is important for testing the trained model correctly.
4. Choose which learner to train on your data: a single learner and choose learner (model) from dropdown menu, or all those currently available.
5. Click "Run pipeline."

When the run starts, the software also checks the input datafile for missing values (NaN, empty cells, etc.).



For now, focus on these outputs and set the rest aside:

- **classification_all** – performance of each classifier, check if it beats the majority-class baseline.
- **ablation_ranking_all** – the smallest number of features needed to maintain optimal accuracy.
- **rankings_label** – the structural features that best distinguish the strains. This is the principal result: it translates the observation that biofilms “look different” into a specific, evidence-based statement of which structural properties differ.
- **rankings_date** – a control. Compared with rankings_label, it identifies features that also predict the imaging date; any feature ranking highly in both is tracking acquisition as much as biology and should be interpreted with caution.

Take home: the model does not simply classify. It produces a ranked, testable account of what differs between groups and flags where that difference may reflect a technical artefact rather than biology.
The remaining output files are described in the homework document and are not required to understand the framework.

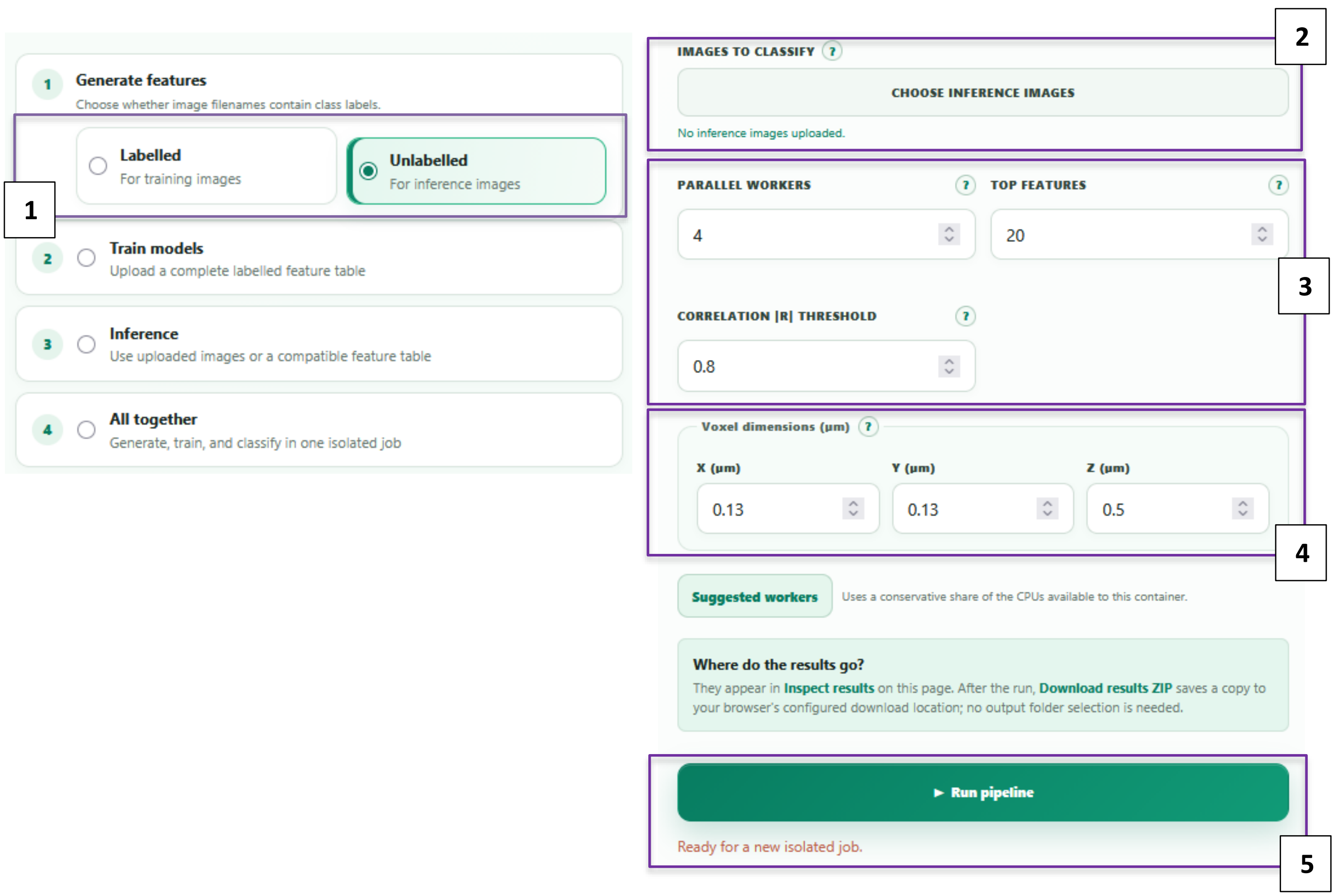
Inference

Inference has two steps: first calculate features from the new images, then run the model on them.

Do step (1) — generate features from the new images:

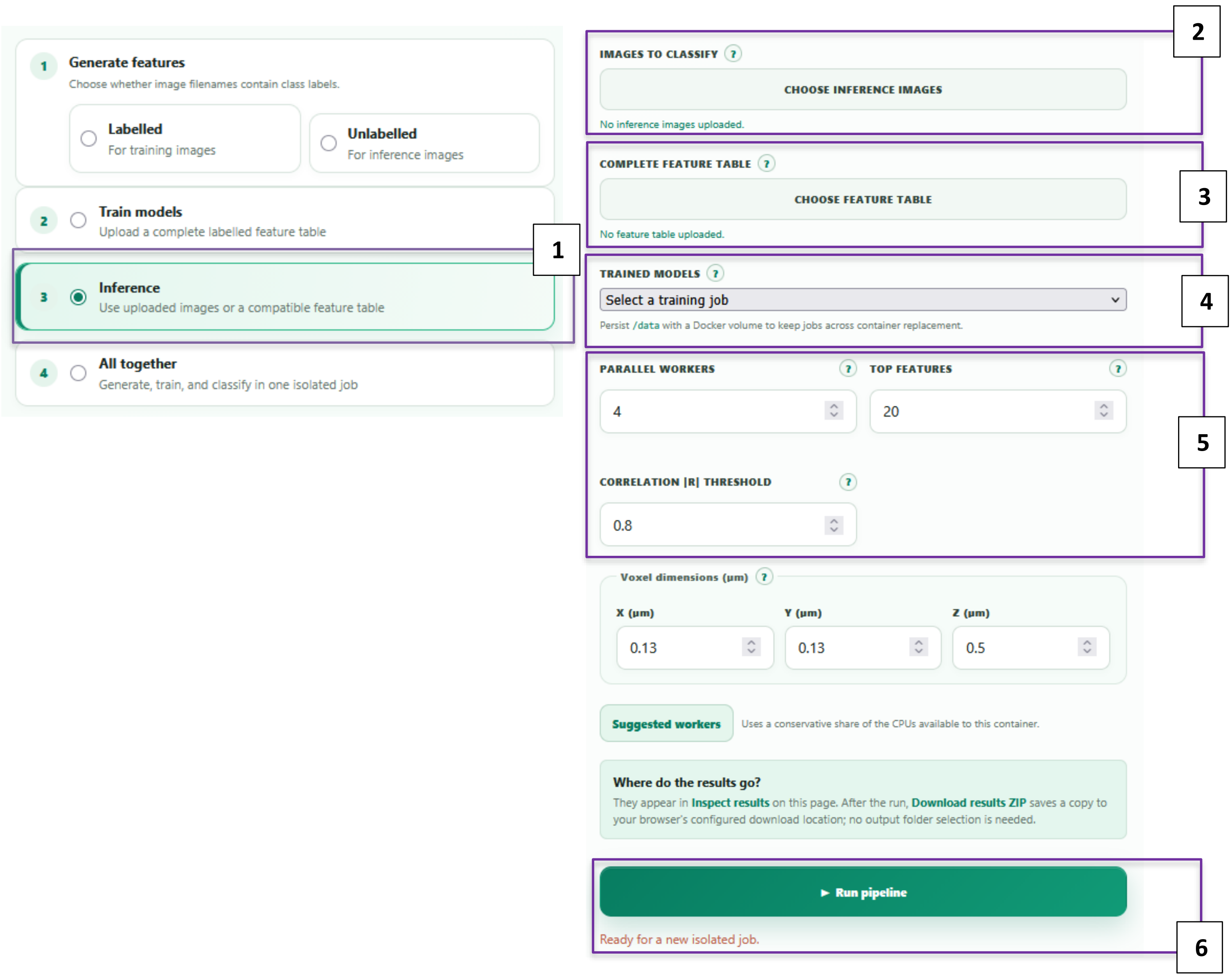
1. Choose "Generate features" in "Unlabelled" mode.
2. Point the software to the folder with the images for inference, using "Choose inference images".
3. Set "Parallel workers" – how much of your CPU to dedicate to the analysis: full capacity when you are not otherwise using the computer, less if you are working on it at the same time.
   "Top features" and "Correlation |R| threshold" do not matter here. Adjust the voxel dimensions (the pixel size in micrometres) to match your images.
4. Click "Run pipeline".

When the run starts, the software checks the image naming but does not read labels – in unlabelled mode the labels are ignored because these images are the unknowns to be classified.



Do step (2) — run the model on those features:

1. Choose “Inference”.
2. Point the software to the folder containing inference images using “Choose inference images”.
3. Point the software to the feature table (the features you just generated in unlabelled mode) using “Choose feature table”.
4. Point the software to the folder containing the trained model.
5. Set “Parallel workers” as before.
6. Click “Run pipeline”.



Read the results, in order:

1. **inference_summary** — sanity check. Strange class names mean the labels weren't read as intended. Stop and fix.
2. **rf_predictions** — the predicted class for each image.
3. **rf_probabilities** — the confidence behind each prediction. A low score is information, not failure: it means the biofilm looks like nothing in the training set.
4. **explanations** — a SHAP plot for your 6 images. Each row is a structural feature; each dot is one of your six images. Dots to the right pushed the model toward a class, dots to the left away; colour is the feature's value (red high, blue low). Read one row to get the idea: if red dots sit right and blue dots left, a higher value of that feature makes a biofilm look more like that class.

Take home: a shifted prediction says the structure moved; the confidence tells you how far and how much to trust it; the explanation says which structural property moved. If that property also matters for distinguishing the strains, it is your hypothesis to test at the bench.

Now it's your turn

- Have fun tonight and let diffuse thinking do the work.
- Tomorrow, read back over what we covered today and highlight anything you did not follow.
- Check the theoretical part to see whether it answers those questions.
- Repeat this exercise.
- Read the paper: https://www.nature.com/articles/s41522-026-01083-8.
- Then move on to the homework, where you run the full dataset and uncover much more detail.